CTMEM-Diff: A Cross-Modal Texture Matching and Enhancement Mamba Diffusion Model for Unpaired Hyperspectral Image Super-Resolution

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***Abstract*—In the field of hyperspectral image super-resolution (HSI-SR), fusion-based image super-resolution (SR) is a research hotspot different from single-image SR (SISR). However, it is common for hyperspectral images (HSIs) and multispectral images (MSIs) to cover different areas and be acquired at different times. Such image pairs pose two challenges to the SR task, which are how to match similar regions of two images and to achieve efficient texture transfer in the complex texture details of remote sensing images. We define this task as unpaired HSI-SR and propose a cross-modal texture matching and enhancement Mamba diffusion (CTMEM-Diff) model to solve the above problems. In the framework of the diffusion model, CTMEM-Diff gradually transfers texture details in MSI in an iterative manner to generate high-resolution HSI (HrHSI). Specifically, at each step of the iterative process, we propose a degradation matching block (DMB), which employs different degradation functions for different modal images to convert HSI and MSI to a uniform data dimension to improve the performance of texture matching. In addition, in order to fully utilize the spectral information in HSI, as well as the spatial information in MSI, we design a texture-consistent Mamba denoising module (TCMM) that utilizes the global context modeling capability and a gating mechanism of Mamba to achieve efficient texture transfer. We constructed three datasets based on five publicly available datasets to validate the effectiveness of our model, and the experimental results show that our method achieves optimal results. The source code is available at https://github.com/Jiahuiqu/CTMEM-Diff**

***Index Terms*—Diffusion model, hyperspectral image (HSI), super-resolution (SR), unpair.**

NOMENCLATURE

Abbreviation	Description
HSI	Hyperspectral image.
MSI	Multispectral image.
LrHSI	Low-resolution HSI.
HrHSI	High-resolution HSI.
HrMSI	High-resolution MSI.
SR	Super-resolution.
SISR	Single-image SR.
Ref-SR	Reference-based image SR.
DDPM	Diffusion model.
CTMEM-Diff	Cross-modal texture matching and enhancement Mamba diffusion model.
TCMM	Texture consistent Mamba denoising module.
DMB	Degradation matching module.
CA	Channel attention.
C^2A	Cross CA.
TEMamba	Texture enhancement Mamba.

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I. INTRODUCTION

HSIs characterize ground objects more finely than RGB images, as different ground objects have different reflective intensities in different wavelength bands. Therefore, HSI has a wide range of applications in various fields, such as change detection [1], [2], ground object classification [3], [4], and semantic segmentation [5], among others. However, HSI has low spatial resolution due to signal-to-noise ratio limitations during imaging, which affects downstream mission applications. Some methods employ spectral SR techniques to reconstruct HSI from readily available RGB images, such as [6], [7], [8], thereby providing data support for downstream tasks. The strengths of these methods lie in the ease of data acquisition and high flexibility, but do not directly address the issue of low resolution in HSI. HSI-SR can improve the spatial resolution of HSI on the software side, thus, it has become a heated topic of research.

HSI-SR can be briefly divided into SISR and fusion-based image SR. Some SISR methods, such as [9], have produced good results. In contrast, fusion-based image SR performs better than SISR because of the auxiliary images providing spatial information. Liu et al. [10] proposed LRTN, decomposing the feature matrix of each channel into a subspace and a space coefficient matrix. By leveraging the spatial low-rank property of HSI, this approach significantly reduces computational complexity. The vast majority of existing fusion-based SR methods are based on the fact that the auxiliary image is a strictly registered MSI of the same area. However, due to the complexity of actual imaging conditions, it is difficult to obtain strictly registered images, which limits the application of fusion-based SR in real-world scenarios. Thus, some works

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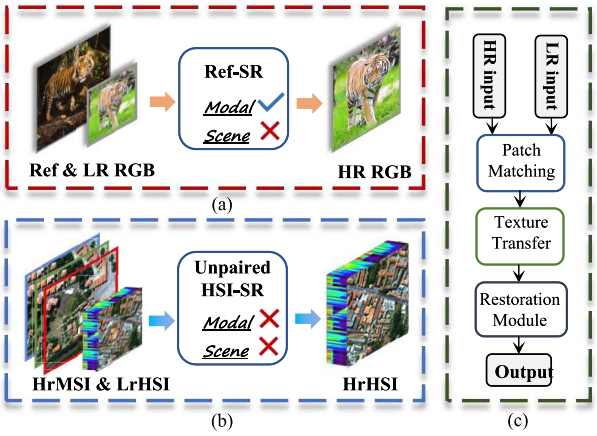


Fig. 1. Paradigms of the Ref-SR and unpaired HSI-SR and the differences between them. (a) Input–output relationship of the Ref-SR method. (b) Input–output relationship of the unpaired HSI-SR method. (c) General procedure of the Ref-SR method.

[11], [12] have gradually emerged to solve the problem of unregistered HSI-SR. In reality, there are still many hyperspectral satellites that are not equipped with MSI sensors, making it impossible to capture high spatial resolution MSIs in both the same time and the same region, which limits the direct application of SR methods based on fused images. For the fusion-based SR method, how to utilize different regions of HrMSI for SR has become a new research hotspot.

There has been precedent in other image SR fields for using high-resolution images of other scenes to aid in SR tasks, which are defined as reference images. The introduction of reference images solves the pathological problem of SISR, where a low-resolution (LR) image corresponds to multiple high-resolution images. These methods are collectively referred to as Ref-SR, where the reference image provides additional semantic, structural, and textural information to the LR image and improves the realism of the generated image, which is consistent with the purpose of fusion-based image SR. [13] uses these high-contrast images (such as 24-month-old images) to guide the segmentation of gray and white matter at six months of age. Some works, such as C^2-matching [14] and AMSA [15], have achieved satisfactory results. However, the limitation of these Ref-SR methods is that their input images are same modality. In terms of HSI-SR, HrHSI of different regions is difficult to obtain, and the SR of LrHSI is usually realized with the help of HrMSI. Existing Ref-SR methods do not take into consideration the data characteristics of HSI as well as the modal differences between images, which weakens the performance of Ref-SR on HSI-SR. We define this task of utilizing HrMSI from different regions to assist in HSI-SR as unpaired HSI-SR, which faces the dual challenge of scene differences and modal differences. The paradigms of the Ref-SR and unpaired HSI-SR are as shown in Fig. 1. The unpaired HSI-SR task differs fundamentally from most fusion-based HSI SR tasks, presenting a more challenging and practical scenario where paired data cannot be obtained.

To address the above problems, we propose a CTMEM-Diff model in this article. The proposed method utilizes the powerful image generation capability of the diffusion model (DDPM) [16]. DDPM generates high-quality images by simulating the diffusion process that restores an image destroyed by noise to a clean image. This way of generating images based on denoising diffusion avoids the instability brought about by discriminator training in generative adversarial networks and provides a better understanding of the process of image generation. Although the U-net denoising network used in the DDPM demonstrated excellent performance, it showed limitations in its ability to model global information. In this aspect, the transformer-based network architecture has shown excellent performance in a number of Ref-SR tasks. The recent emergence of the Mamba architecture [17] has posed a significant challenge to transformer-based models' longstanding dominance in global contextual modeling capabilities, particularly regarding their effectiveness in capturing long-range dependencies. Many Mamba-based models [18], [19], [20] prove their potential in HSI-SR, especially fusion-based image SR. Therefore, we design a TCMM to replace the U-net denoising module in traditional DDPM to better model the global information. Considering the data characteristics of HSI, we add the attention mechanism for the channel direction before Mamba, because Mamba has an extremely high computational cost to scan in the spectral direction.

In addition, for the problem of modal and scene differences between HSI and MSI, we designed a degradation matching block (DMB), which uses different degradation functions for different modal images to convert different modal images to a unified data dimension, thus improving the matching effectiveness.

The main contributions of this article are summarized as follows.

1) We propose a CTMEM-Diff model, which achieves excellent results on unpaired HSI-SR tasks. To the best of our knowledge, we are one of the first to research the task of unpaired HSI-SR.
2) We design a DMB, whose core is to unify the data dimensions of different modal images by different degenerate functions to achieve more effective texture matching.
3) We propose a TCMM, which effectively utilizes the spatial–spectral information to achieve better reconstruction results through the global context modeling capability and a gating mechanism of Mamba.

The rest of the article is organized as follows. In Section II, we review some related works. Section III describes the proposed CTMEM-Diff. In Section IV, we give qualitative and quantitative results of the proposed method on three synthetic datasets with ablation studies. The abbreviations used in this article and their definitions are shown in Nomenclature.

II. Related Work

In this section, we review some representative work on two classes of methods, fusion-based image SR and Ref-SR, to describe the current status of research.

A. Fusion-Based Image SR

Different from SISR, fusion-based image SR methods have better fidelity of SR results because of the intervention of

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auxiliary images. Early fusion-based SR methods were built on end-to-end networks to learn mapping relationships from LrHSI and HrMSI to HrHSI. Zhang et al. [21] proposed an adaptive HSI-MSI learning framework to fully utilize the a priori information of HSI and MSI. Combined with the guidance of effective mathematical theory, Dong et al. [22] designed a dual spatial–spectral optimization strategy and introduced two optimization branches based on spatial and spectral priors, respectively. In order to solve the problem that CNN cannot utilize global information, many transformer-based HSI-SR methods have also been proposed. Wang et al. [23] proposed a transformer-based network that utilizes cross-attention for information fusion and achieves multilevel feature extraction and aggregation. For the problem of unregistered HSI-SR, Qu et al. [24] used cross-attention for capturing spatial correlations at long distances to repair nonrigid pixel offsets and achieved the registration-fusion co-optimization. Recently, some methods based on Mamba have been proposed to replace the transformer. Panmamba [18] uses channel switching to fuse information based on Mamba. Zhao et al. [19] proposed a space-spectral interaction-guided fusion network combining CNN and Mamba to efficiently extract local–global features through mutual guidance to improve resolution.

The diffusion model has emerged as an innovative method that iteratively refines the noise vector into the desired output. This reversible process generates high-quality samples and has been demonstrated in various applications. The conditional diffusion model SR3 proposed by Saharia et al. [25] is widely used with SR tasks, which is guided by the up-sampling LR image, and generates many reliable details in the results. Meanwhile, the diffusion model has also been applied to HSI-SR. [26] used the powerful image modeling capabilities of the transformer to replace the denoising process with a sampling process of the noisy image, making SR results more reliable. [27] proposed a conditional circular diffusion model structure that produces HrHSI with complementary spatial and spectral information through SR in the spatial and spectral domains.

B. Reference-Based Image SR

The purpose of Ref-SR is to transform an LR image into a high-resolution image by utilizing the texture information in the reference image. In order to accomplish the goal, it is necessary to match the content of LR images with that of high-resolution images. Zheng et al. [28] used crossnet to warp reference features at multiple scales to align reference features with LR. Shim et al. [29] extracted and aligned reference features using deformable convolution. However, for most cases, the features of the reference image and the LR cannot be aligned by offsetting the pixels. As a derivation method of pixel shifting, patch matching has been widely used for feature matching in reference and LR images. Early works such as SRTNN [30] use pretrained VGG networks for content similarity patch matching and feature migration. TTSR [31] stacks multiple proposed texture transformers on multiple scales to discover different levels of texture. Cao et al. [32] proposed a multiscale deformable attention transformer (DATSR) to compensate low-score images using multiple relevant textures. Jiang et al. [14] combined contrastive learning and teacher–student correlation distillation with pixel matching to narrow matching differences. Wang et al. [33] employed a progressive matching mode to effectively match favorable high-frequency information at each resolution.

The above-mentioned Ref-SR methods have achieved satisfactory results in the field of natural images; meanwhile, some methods have recently been applied to remote sensing images as well. Dong et al. [34] used the change detection results as a priori for guiding the SR of LR. RRSGAN [35] uses a gradient-assisted feature alignment method that uses deformable convolution to align Ref and LR features, and a relevant attention module (RAM) to improve the robustness of the model in different scenarios. RGTGAN [36] uses gradient-assisted texture enhancement to and dense-internal deformable convolution to improve feature alignment.

These Ref-SR methods applied to remote sensing images are limited to RGB images with different phases or viewpoints in the same region, and have not been investigated in the situation where the scene differences are much larger and the modal differences exist between HSI and MSI.

III. Proposed Method

In this section, we describe in detail the problem addressed by the method proposed in this article, and the way it is implemented. As shown in Fig. 2, our method chooses the diffusion model as the base framework and changes the denoising network and denoising method for the reverse denoising process. The reverse denoising network consists of two parts, the DMB and the TCMM. The DMB realizes the texture matching between the reference HrMSI and the LrHSI and extracts the texture features; the TCMM transfers the extracted texture features to the LrHSI to generate the desired HrHSI. In the following, we will describe the above in detail.

A. Problem Formulation

Given LrHSI $\mathbf{X} \in \mathbb{R}^{C\times h\times w}$ and unpaired HrMSI $\mathbf{Y}^{\text{ref}} \in \mathbb{R}^{c\times H\times W}$, where $\{C, c\}$ denote the number of bands, $\{H, h\}$, $\{W, w\}$ denote height, and $\{W, w\}$ denote the width of the images, respectively, and where H, W, and C are larger than h, w, and c, their relationship to HrHSI is consistent with the observation model as shown in the following equation:

$$\begin{aligned}\mathbf{X} &= \mathbf{ZB}\\ \mathbf{Y}^{\text{ref}} &= \mathbf{RZ}^{\text{ref}}\end{aligned} \tag{1}$$

where $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$ denotes the HrHSI corresponding to the $\mathbf{X}$, and $\mathbf{Z}^{\text{ref}} \in \mathbb{R}^{C\times H\times W}$ denotes the HrHSI corresponding to the unpaired HrMSI $\mathbf{Y}^{\text{ref}}$, and $\mathbf{B}$ denotes spatial degradation matrix, and $\mathbf{R}$ denotes the spectral response function. Estimating $\mathbf{Z}$ solely from $\mathbf{X}$ is inherently ill-posed due to the nonunique mapping, where any single $\mathbf{X}$ can correspond to multiple possible solutions for $\mathbf{Z}$. Consequently, traditional fusion methods enhance the performance by introducing auxiliary images, which conform to the following paradigm:

$$\{\mathbf{X}, \mathbf{Y}\} \xrightarrow{f} \mathbf{Z} \tag{2}$$

where $\mathbf{Y} \in \mathbb{R}^{c\times H\times W}$ denotes the HrMSI in the same region with the same temporal phase as $\mathbf{X}$, and f denotes the function

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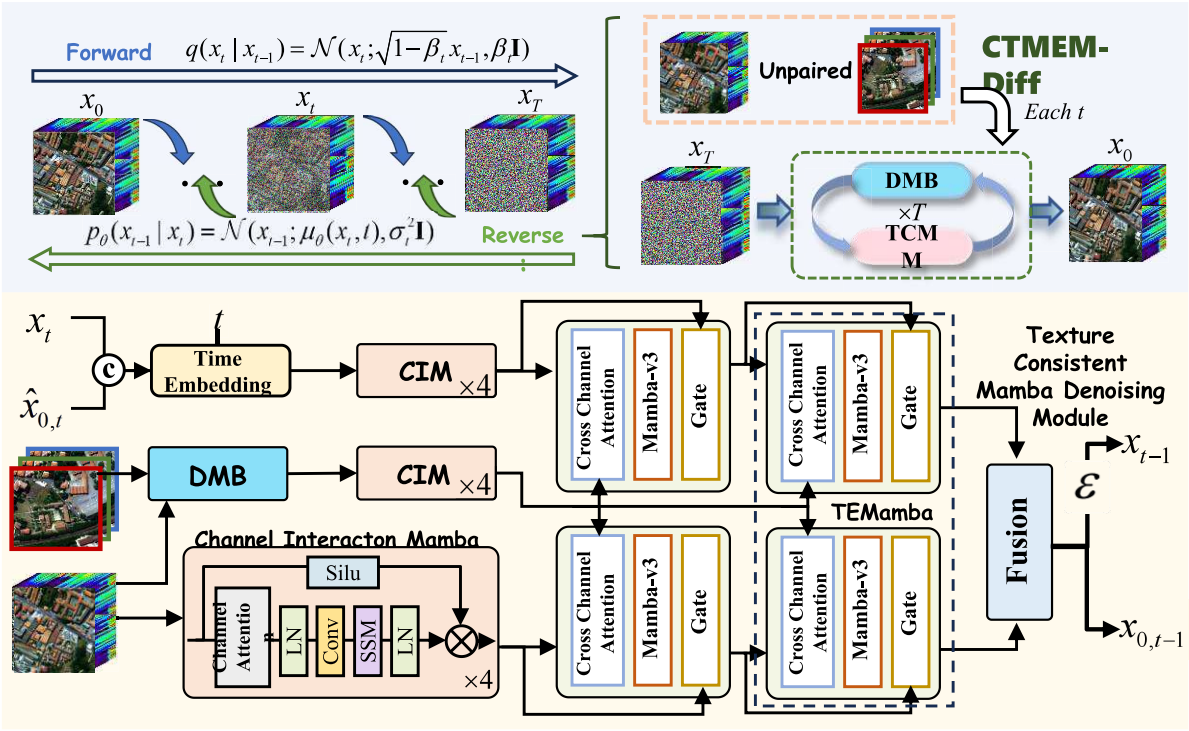


Fig. 2. (Top left) Flowchart of the reverse texture matching-enhancement process of the diffusion model. (Top right) Overall flowchart of the CTMEM-Diff, and the bottom is the detailed flowchart of one step in the reverse texture matching-enhancement process.

that optimally maps $\mathbf{X}$ and $\mathbf{Y}$ to $\mathbf{Z}$. The mutual constraints between $\mathbf{X}$ and $\mathbf{Y}$ guarantee the uniqueness of $\mathbf{Z}$.

In general, strictly registered LrHSI and HrMSI are difficult to obtain. Similarly, we use $\{\mathbf{X}, \mathbf{Y}^{\text{ref}}\}$ as model inputs to deal with the absence of paired images. The traditional assumption for unpaired image SR is that the input image modalities are identical. The significant modal differences between HSI and MSI, especially in channel numbers, pose challenges for texture matching and transfer. To address the challenges of scene inconsistency and modality differences in fusion-based image SR, we propose an unpaired HSI-SR method to learn the optimal mapping in the following equation:

$$\{\mathbf{X}, \mathbf{Y}^{\text{ref}}\} \xrightarrow[\mathcal{M}]{\mathcal{F}} \mathbf{Z} \tag{3}$$

where $\mathcal{M}$ represents the texture matching, and $\mathcal{F}$ denotes reconstruction process based on matching textures.

B. Overall Pipeline

In this article, we propose the CTMEM-Diff model, which can realize the effect of (3). Considering the problem of scene inconsistency between $\mathbf{X}$ and $\mathbf{Y}^{\text{ref}}$, the proposed method matches $\mathbf{X}$ and $\mathbf{Y}^{\text{ref}}$ by patch-matching to obtain the texture information and transfers it from $\mathbf{Y}^{\text{ref}}$ to $\mathbf{X}$ in an iterative manner, so that $\mathbf{X}$ obtains the desired texture details and finally the desired $\mathbf{Z}$. The core problem addressed in this method is how to realize the texture matching between $\mathbf{X}$ and $\mathbf{Y}^{\text{ref}}$, and how to utilize the matched texture information to realize the texture enhancement of $\mathbf{X}$. The proposed framework is shown in Fig. 2, which general architecture is the diffusion model.

During the forward process, Gaussian noise is gradually added to $\mathbf{Z}$ until it is completely destroyed as pure Gaussian noise $\mathbf{Z}_T$. In the reverse texture matching-enhancement process, under the conditions of $\mathbf{X}$ and $\mathbf{Y}^{\text{ref}}$, the network gradually transforms $\mathbf{Z}_T$ to $\mathbf{Z}_0$. The network we designed for the inverse process consists of two main structures, DMB and TCMM. The former achieves effective texture matching, and the latter transfers the corresponding texture information to LR images to enhance image quality. DMB employs different degradation processes to make $\mathbf{X}$ and $\mathbf{Y}^{\text{ref}}$ have similar forms of data representation, which makes the process of similarity matching more efficient. In addition, the matching results contain the original image patches of $\mathbf{Y}^{\text{ref}}$, but DMB also adds gradient image patches and residual image patches, as well as image patches weighted with attention scores. Subsequently, TCMM exploits the capability of modeling contextual relationships with the Mamba structure to better accomplish the task of texture transfer and enhancement. At each step of the reverse process, the designed DMB actively performs texture matching between unpaired LrHSI and HrMSI at every iteration, enabling the TCMM to extract targeted texture information from the matching results.

In contrast to conventional diffusion paradigms that estimate iterative noise perturbations, the proposed TCMM directly predicts the original image $\mathbf{Z}_0$. While this approach limits the generative diversity inherent in the diffusion process, it significantly improves reconstruction fidelity by enforcing spectral spatial consistency—a key requirement for remote sensing applications, where the reality of the geographic information is more important than the diversity. Building upon this,

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the intermediate prediction $\mathbf{Z}_{0,t}$ and denoised result $\mathbf{Z}_t$ from the current step are jointly propagated into the subsequent network, which is described in Section III-D. After the gradual iteration of the inverse process, the texture information of the generated image is gradually enriched, and finally, the desired HrHSI is obtained.

C. Forward Diffusion Process

The forward process adds Gaussian noise to $\mathbf{Z}_0$ over T iterations, which is shown as follows:

$$\begin{aligned} q\left(\mathbf{Z}_{1:T}|\mathbf{Z}_0\right) &= \prod_{t=1}^{T} q\left(\mathbf{Z}_t|\mathbf{Z}_{t-1}\right) \\ q\left(\mathbf{Z}_t|\mathbf{Z}_{t-1}\right) &= \mathcal{N}\left(\mathbf{Z}_t;\ \sqrt{\alpha_t}\mathbf{Z}_{t-1}, (1-\alpha_t)\,\mathbf{I}\right) \end{aligned} \tag{4}$$

where $\alpha_{1:T} \in (0,1)$ is a known hyperparameter. According to the reparameterization trick, $\mathbf{Z}_t$ can be generated directly from $\mathbf{Z}_0$

$$\begin{aligned} \mathbf{Z}_t &= \sqrt{\tilde{\alpha}_t}\mathbf{Z}_0 + \sqrt{1-\tilde{\alpha}_t}\epsilon \\ q\left(\mathbf{Z}_t|\mathbf{Z}_0\right) &= \mathcal{N}\left(\mathbf{Z}_t;\ \sqrt{\tilde{\alpha}_t}\mathbf{Z}_0, (1-\tilde{\alpha}_t)\,\mathbf{I}\right) \end{aligned} \tag{5}$$

where $\tilde{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. Given $\mathbf{Z}_0$ and $\mathbf{Z}_t$, the posterior distribution of $\mathbf{Z}_{t-1}$ can be computed be computed as follows:

$$\begin{aligned} q\left(\mathbf{Z}_{t-1}|\mathbf{Z}_0, \mathbf{Z}_t\right) &= \mathcal{N}\left(\mathbf{Z}_{t-1}; \mu, \sigma^2\mathbf{I}\right) \\ \mu &= \frac{\sqrt{\tilde{\alpha}_{t-1}}\,(1-\alpha_t)}{1-\tilde{\alpha}_t}\mathbf{Z}_0 + \frac{\sqrt{\alpha_t}\,(1-\tilde{\alpha}_{t-1})}{1-\tilde{\alpha}_t}\mathbf{Z}_t \\ \sigma^2 &= \frac{(1-\tilde{\alpha}_{t-1})\,(1-\alpha_t)}{1-\tilde{\alpha}_t}. \end{aligned} \tag{6}$$

D. Reverse Texture Matching-Enhancement Process

The purpose of the reverse process is to progressively remove the Gaussian noise added in the forward process and obtain a pure $\mathbf{Z}_0$. Guided by the framework of the conditional diffusion model, the information in $\mathbf{X}^L$ and $\mathbf{Y}^H$ can be rationally utilized in each iteration step, and the process is shown as follows:

$$\begin{aligned} p_\theta\left(\mathbf{Z}_{0:T}|\mathbf{X}, \mathbf{Y}^{\text{ref}}\right) &= p\left(\mathbf{Z}_\mathbf{T}\right) \prod_{t=1}^{T} p_\theta\left(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{X}, \mathbf{Y}^{\text{ref}}\right) \\ p\left(\mathbf{Z}_T\right) &= \mathcal{N}\left(\mathbf{Z}_T; 0, \mathbf{I}\right) \\ p_\theta\left(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{X}, \mathbf{Y}^{\text{ref}}\right) &= \mathcal{N}\left(\mathbf{Z}_{t-1}; \mu_\theta\left(\mathbf{X}, \mathbf{Y}^{\text{ref}}, \mathbf{Z}_\mathbf{t}, \tilde{\alpha}_t\right), \sigma_t^2\mathbf{I}\right). \end{aligned} \tag{7}$$

According to (6), CTMEM-Diff obtains the mean μ_θ in a way that predicts $\widetilde{\mathbf{Z}}_0$. In the tth step iteration, the predicted $\widetilde{\mathbf{Z}}_0$ is denoted as $\mathbf{Z}_{0,t-1}$. Each iteration step is represented as follows:

$$\begin{aligned} \mathbf{Z}_{0,t-1} &= g_\theta\left(\mathbf{Z}_t, \mathbf{Z}_{0,t}, \mathbf{X}, \mathbf{Y}^{\text{ref}}, \tilde{\alpha}_t\right) \\ \mathbf{Z}_{t-1} &= \frac{\sqrt{\tilde{\alpha}_{t-1}}\,(1-\alpha_t)}{1-\tilde{\alpha}_t}\mathbf{Z}_{0,t-1} \\ &\quad + \frac{\sqrt{\alpha_t}\,(1-\tilde{\alpha}_{t-1})}{1-\tilde{\alpha}_t}\mathbf{Z}_t + \sqrt{1-\tilde{\alpha}_t}\epsilon \end{aligned} \tag{8}$$

where $g_\theta(\cdot)$ denotes the denoising network, and $\mathbf{Z}_{0,t}$ denotes the prediction of $\widetilde{\mathbf{Z}}_0$ used to acquire the noisy image $\mathbf{Z}_t$. Note that we believe that the predicted $\mathbf{Z}_{0,t}$ contains some additional information, so we connect it to $\mathbf{Z}_t$ and use it as a basis for the next iteration. Disregarding the first step of the iteration, the distribution $p_\theta(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{X}^L, \mathbf{Y}^H)$ of (7) can be rewritten as

$$\begin{aligned} &p_\theta\left(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{Z}_{0,t}, \mathbf{X}, \mathbf{Y}^{\text{ref}}\right) \\ &= \mathcal{N}\left(\mathbf{Z}_{t-1}; \mu_\theta\left(\mathbf{Z_t}, \mathbf{Z}_{0,t}, \mathbf{X}, \mathbf{Y}^{\text{ref}}, \tilde{\alpha}_t\right), \sigma_t^2\mathbf{I}\right). \end{aligned} \tag{9}$$

Specifically, the reverse texture matching-enhancement process consists of two procedures. First, $\mathbf{Y}^H$ and $\mathbf{X}^L$ are texture-matched in DMB to get the texture matching information map $\mathcal{M}^H$, and then $\mathcal{M}^H$ with $\mathbf{X}^L$ and noise image is fed into TCMM for texture transfer and enhancement. Each step of the reverse process can be expressed as follows:

$$\begin{aligned} \mathcal{M}^H &= \text{DMB}\left(\mathbf{X}, \mathbf{Y}^{\text{ref}}\right) \\ \mathbf{Z}_{0,t-1} &= \text{TCMM}\left(\mathbf{Z}_t, \mathbf{Z}_{0,t}, \mathcal{M}^H, \tilde{\alpha}_t\right) \\ \mathbf{Z}_{t-1} &= \mathbf{Z}_{0,t-1} + \sqrt{1-\tilde{\alpha}_t}\epsilon \end{aligned} \tag{10}$$

where TCMM$(\cdot)$ denotes TCMM. We describe DMB and TCMM in detail in Sections III-D1 and III-D2.

1) Degradation Matching Block: As mentioned earlier, optical images of different modalities pose a challenge for texture matching. To solve this problem, we propose the DMB, which transforms $\mathbf{X}$ and $\mathbf{Y}^{\text{ref}}$ into uniform data dimensions by means of degenerate estimation for better texture matching processing. In a step of the reverse texture matching-enhancement process, $\mathbf{Y}^{\text{ref}}$ undergoes spatial degradation, and $\mathbf{X}$ undergoes spectral degradation, respectively. This step is represented as follows:

$$\begin{aligned} \mathbf{F}^H &= f_{\text{sd}}\left(\mathbf{Y}^{\text{ref}}\right) \\ \mathbf{F}^L &= f_{\text{rd}}\left(\mathbf{X}\right) \end{aligned} \tag{11}$$

where $f_{\text{sd}}(\cdot)$ and $f_{\text{rd}}(\cdot)$ denote the spatial degradation module and the spectral degradation module, respectively, and $\mathbf{F}^H$ and $\mathbf{F}^L$ denote MSI degradation features and HSI degradation features, respectively. $f_{\text{sd}}(\cdot)$ is implemented using an encoder, a series of convolutional kernels of varying sizes, and a decoder. $f_{\text{sd}}(\cdot)$ simulates the difference in spatial information that exists between $\mathbf{Y}^{\text{ref}}$ and $\mathbf{X}$ in a manner that degrades the information in a latent space. $f_{\text{rd}}(\cdot)$realizes spectral degradation as a spectral response function. As depicted in Fig. 3, spectral

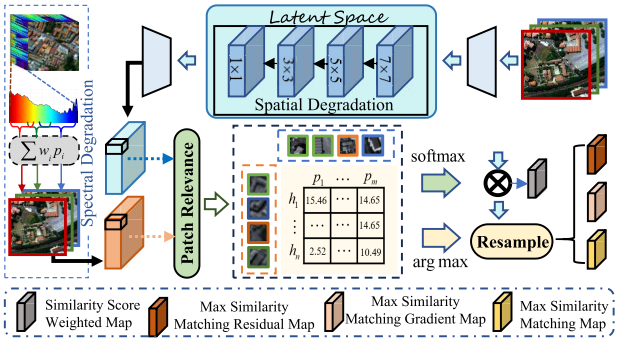


Fig. 3. Structure of the DMB. MSI reduces the spatial resolution through latent space degradation, and the degraded part is composed of multiple convolution kernels of different sizes. HSI undergoes spectral degradation through the spectral response mechanism.

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degradation is achieved by generating an MSI through the discrete, weighted integration of specific narrowband wavelengths (e.g., blue, green, red, and infrared) from the HSI.

Subsequently, the degraded feature maps, $\mathbf{F}^H$ and $\mathbf{F}^L$, are partitioned into N_h and N_l nonoverlapping patches, respectively. We then compute the patch-wise similarity matrix $\mathbf{S}$ using cosine similarity to effectively capture correlations between textures in different patches.

To optimally leverage textural information, we compute and apply softmax normalization to patch-wise similarity scores $\mathbf{S}$ to derive a similarity matrix, which assigns higher weights to more correlated patches. These weights are then utilized for adaptive aggregation generating a similarity-weighted feature map. The process is illustrated as follows:

$$\mathbf{M}_{\text{sw}} = \text{Softmax}\left(\mathbf{S}\right) \cdot \mathcal{P}\left(\mathbf{F}^H\right) \tag{12}$$

where $\mathbf{M}_{\text{sw}}$ denotes the similarity score weighted map, $\mathbf{S} \in \mathbb{R}^{N_l \times N_h}$ denotes the patch-wise similarity scores, and $\mathcal{P}(\mathbf{F}^H) \in \mathbb{R}^{N_h \times 1}$ denotes the sequence of nonoverlapping patches partitioned by $\mathbf{F}^H$.

Meanwhile, we compute a hard attention map $\mathbf{H}$ where the ith element h_i is computed as follows:

$$h_i = \arg\max_i \left(\mathbf{S}_{i,j}\right). \tag{13}$$

According to the computed hard attention map $\mathbf{H}$, we resample $\mathbf{Y}^H$, the gradient map of $\mathbf{Y}^H$, and the residual map of $\mathbf{Y}^H$. The residual map is obtained by calculating the difference between $\mathbf{Y}^H$ and its degraded version, while the gradient map is generated by computing the normalized gradient of each pixel in $\mathbf{Y}^H$. The original image has richer detail information, while the residual and gradient maps are rich in high-frequency information. Different texture matching maps are concatenated in the channel direction to obtain $\mathcal{M}^H$.

2) Texture Consistent Mamba Denoising Module: In order to better transfer the texture information in $\mathcal{M}^H$ into HSI, we propose TCMM for replacing the U-net denoising network, and the specific architecture is shown in Fig. 2. TCMM mainly consists of three encoder branches and one decoder branch. According to the characteristics of HSI, the CIM that makes up the encoder adds the CA block in front of the Mamba module to calculate the context relationship of the channel direction, and its computational complexity is much lower than that of the Mamba block of channel direction scanning. The computational formulas of CIM are shown as follows:

$$\begin{aligned}
\mathbf{F}' &= \text{CA}(\mathbf{F}_{\text{in}}) \\
\mathbf{T}' &= \text{Linear}(\text{SSM}(\text{Conv}(\text{Linear}(\mathbf{F}')))) \\
\mathbf{T}_{\text{out}} &= \mathbf{T}' * \text{Silu}(\mathbf{F}_{\text{in}})
\end{aligned} \tag{14}$$

where CA(·) denotes the CA, SSM(·) denotes state space model, Conv(·) denotes convolution operation, Silu(·) denotes activate function, Linear(·) denotes linear projection, $\mathbf{T}'$ and $\mathbf{F}'$ denote intermediate features, $\mathbf{F}_{\text{in}}$ denotes the input of the first CIM, and $\mathbf{T}_{\text{out}}$ denotes the output of the first CIM. The process of calculating CA is as follows:

$$\begin{aligned}
&\mathbf{Q} = W_q\left(\mathbf{I}\right), \quad \mathbf{K} = W_k\left(\mathbf{I}\right), \quad \mathbf{V} = W_v\left(\mathbf{I}\right) \\
&\mathbf{O} = \text{Attention}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) + \mathbf{I}
\end{aligned}$$

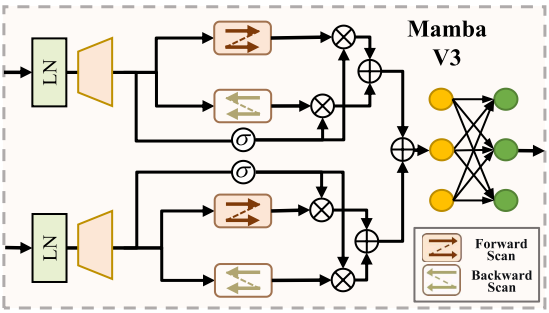


Fig. 4. Structure of Mamba V3. LN represents LayerNorm, the $\otimes$ represents the multiplication of main elements, the $\oplus$ represents element-wise addition, and σ symbol represents the activation function.

$$\text{Attention}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) = \text{Softmax}\left(\mathbf{Q} \cdot \mathbf{K}^T\right) \cdot \mathbf{V} \tag{15}$$

where $\mathbf{I} \in \mathbb{R}^{c \times \text{hw}}$ denotes the input of CA(·), $\mathbf{O} \in \mathbb{R}^{c \times \text{hw}}$ denotes the output of CA(·), and $\mathbf{Q} \in \mathbb{R}^{c \times \text{hw}}$, $\mathbf{K} \in \mathbb{R}^{c \times \text{hw}}$, $\mathbf{V} \in \mathbb{R}^{c \times \text{hw}}$ are the projections of $\mathbf{I}$. Here, c, h, and w refer to the number of channels, height, and width of the input CA(·) module features, which is different from the number of channels, height, and width of the image mentioned earlier. This data dimension in CA has to be transformed into the dimension of $\mathbf{Q} \cdot \mathbf{K}^T$ as $C \times C$, and the validity of this way of computing correlations along the direction of the channel is confirmed in [37].

The decoder branch of TCMM consists of the TEMamba, whose internal structure consists of a C^2A, a Mamba V3, and a gate module. Furthermore, in the decoder branch of TCMM, the features of $\mathcal{M}^H$ are transferred to the features of $\mathbf{X}^L$ and $\mathbf{Z}_t$ by C^2A and Mamba v3. The computation of C^2A involves projecting texture features into $\mathbf{Q}$ and the features of another branch into $\mathbf{K}$ and $\mathbf{V}$. Channel-wise attention is performed between $\mathbf{Q}$ and $\mathbf{V}$, resulting in a similarity map, which is then multiplied with $\mathbf{V}$ to produce the final output. This process can be expressed as follows:

$$\begin{aligned}
\mathbf{Q}_t = W_q\left(\mathbf{F}_t\right), \ \mathbf{K}_f = W_k\left(\mathbf{F}_f\right), \ \mathbf{V}_f = W_v\left(\mathbf{F}_f\right) \\
\text{Attention}\left(\mathbf{Q}_t, \mathbf{K}_f, \mathbf{V}_f\right) = \text{Softmax}\left(\mathbf{Q}_t \cdot \mathbf{K}_f^T\right) \cdot \mathbf{V}_f \\
\mathbf{O}_f = \text{Attention}\left(\mathbf{Q}_t, \mathbf{K}_f, \mathbf{V}_f\right) + \mathbf{F}_f
\end{aligned} \tag{16}$$

where $\mathbf{F}_t$ represents the texture feature, $\mathbf{F}_f$ represents the feature of another branch, W_q, W_k, and W_v represent the projection matrices, $\mathbf{Q}_t$, $\mathbf{K}_f$, and $\mathbf{V}_f$ represent the corresponding projected features, and $\mathbf{O}_f$ represents the output feature. The structure of the TEMamba is shown in Fig. 2 and the structure of Mamba v3 is shown in Fig. 4. Finally, the features of the two branches are passed through a fusion module to obtain the final predicted $\mathbf{Z}_{0,t-1}$, and $\mathbf{Z}_{t-1}$ is computed from (8). The complete TCMM can be expressed as

$$\begin{aligned}
\mathbf{Z}_{0,t-1} &= \text{TCMM}\left(\mathbf{Z}_t, \mathbf{Z}_{0,t}, \mathcal{M}^H, \mathbf{X}, \tilde{\alpha}_t\right) \\
\mathbf{Z}_{t-1} &= \mathbf{Z}_{0,t-1} + \sqrt{1-\tilde{\alpha}_t}\epsilon
\end{aligned} \tag{17}$$

where TCMM(·) denotes TCMM. In the input of the encoder branch of the noise of the TCMM, we add the result $\mathbf{Z}_{0,t}$ predicted in the previous step, which helps to get information from the noise about the details that were not recovered by the denoising process in the previous step.

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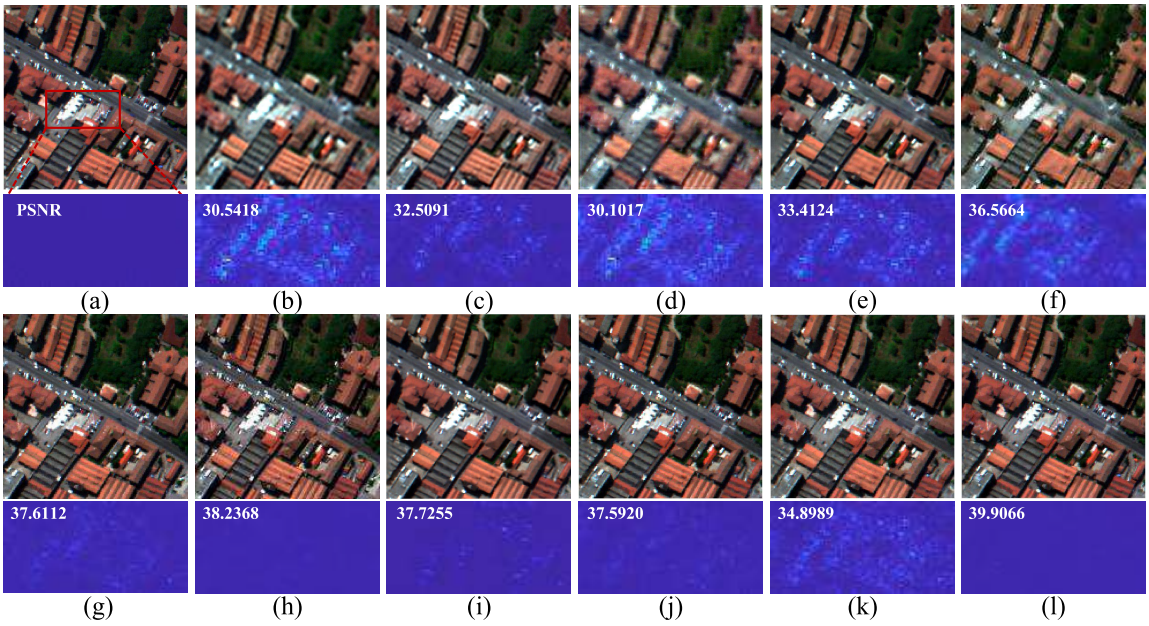


Fig. 5. Visual results and MAE maps generated by different HSI-SR algorithms for the unpaired Pavia dataset. (a) Reference. (b) MSDFormer. (c) RepMamba. (d) CLSCNet. (e) Aware. (f) Aware (DMB). (g) Panmamba. (h) Panmamba (DMB). (i) MSAN. (j) DATSR. (k) RGTGAN. (l) Ours.

It is worth noting that we have added a number of gating modules in the encoder and decoder to ensure that texture enhancement can be achieved while maintaining the original data distribution. The gate module can be represented as follows:

$$\mathbf{F}_{\text{out}} = \text{ReLU}\left(\text{Conv}\left(\mathbf{F}_{\text{in}}^{1}\right)\right) \cdot \text{Conv}\left(\mathbf{F}_{\text{in}}^{2}\right)) + \mathbf{F}_{\text{in}}^{1} \tag{18}$$

where $\mathbf{F}_{\text{out}}$ denotes the output features of gate module, $\mathbf{F}_{\text{in}}^{1}$ and $\mathbf{F}_{\text{in}}^{2}$ denotes the inputs of the gate module, Conv denotes the convolution operation, and ReLU denotes the activation function.

Although the matching process only extracts similar patches and does not perform pixel-level alignment, the global information extraction capabilities of the transformer and Mamba enable the network to utilize texture information from a large field of view, thereby improving texture aggregation capabilities.

E. Loss Function

Consistent with other HSI-SR methods, we choose L1 loss as our loss function. The loss function is expressed as

$$\mathcal{L} = \|\mathbf{Z}_{0,t} - \mathbf{Z}\|_1 \tag{19}$$

where $\mathbf{Z}_{0,t}$ denotes the output of the network in the reverse texture matching-enhancement process, and $\mathbf{Z}$ denotes the ground truth.

IV. Experiment Results and Analysis

A. Dataset

To verify the effectiveness of the proposed method, we conduct experiments on three datasets, which are constructed based on five publicly available datasets. The HrMSI and LrHSI used in Pavia, Houston, and ChikuseiR datasets are generated according to Wald's protocol.

1) Pavia Dataset: The Pavia dataset splits the Pavia Center dataset into 12 patches of size $160 \times 320 \times 102$, 9 for training and 3 for testing. For each patch, we cut it into 21 patches of size $160 \times 160 \times 102$ with a step size of 8 as the final HrHSI, and its corresponding LrHSI size is $40 \times 40 \times 102$. Moreover, we replaced the MSI of the Pavia Center dataset with the MSI of the Pavia University dataset. In order to cover all the ground object information, we take the unsliced HrMSI of the Pavia University dataset of size $640 \times 320 \times 4$ as input.

2) Houston Dataset: The Houston dataset was constructed in the same manner as the Pavia dataset. We replaced the MSI in the Houston 2018 dataset with the MSI of the Houston 2013 dataset. To align the spatial resolution of HSI versus MSI, we sampled the Houston 2018 dataset in the spatial dimension by a 1/2 scale factor. We intercepted images of size $480 \times 1280 \times 48$ for experiments, which were sliced into 8 patches of size $480 \times 160 \times 48$, 6 for training and 2 for testing. The sizes of LrHSI, HrHSI, and HrMSI in the Houston dataset are $40 \times 40 \times 48$, $160 \times 160 \times 48$, and $320 \times 1120 \times 4$, respectively.

3) ChikuseiR Dataset: The ChikuseiR dataset uses a region of size $2048 \times 2048 \times 128$ after removing edges from the original Chikusei dataset [38]. To achieve the unpaired effect, it is cut into left and right parts of size $2048 \times 1024 \times 128$, and the HSI of the left part and the MSI of the right part are removed. We cut the right part into 16 patches of $512 \times 256 \times 128$, 12 for training and 4 for testing. Furthermore, these patches are sliced in steps of 32 into smaller patches of $256 \times 256 \times 128$ as the HrHSI of the dataset. The size of the LrHSI corresponding to the HrHSI is $64 \times 64 \times 128$.

4) Liyukou Dataset: The Liyukou dataset covers an area in the Huyi District of Xi'an, Shaanxi Province, and includes an HSI captured on April 6, 2023, and an RGB image captured on November 18, 2022. The RGB image was acquired using a

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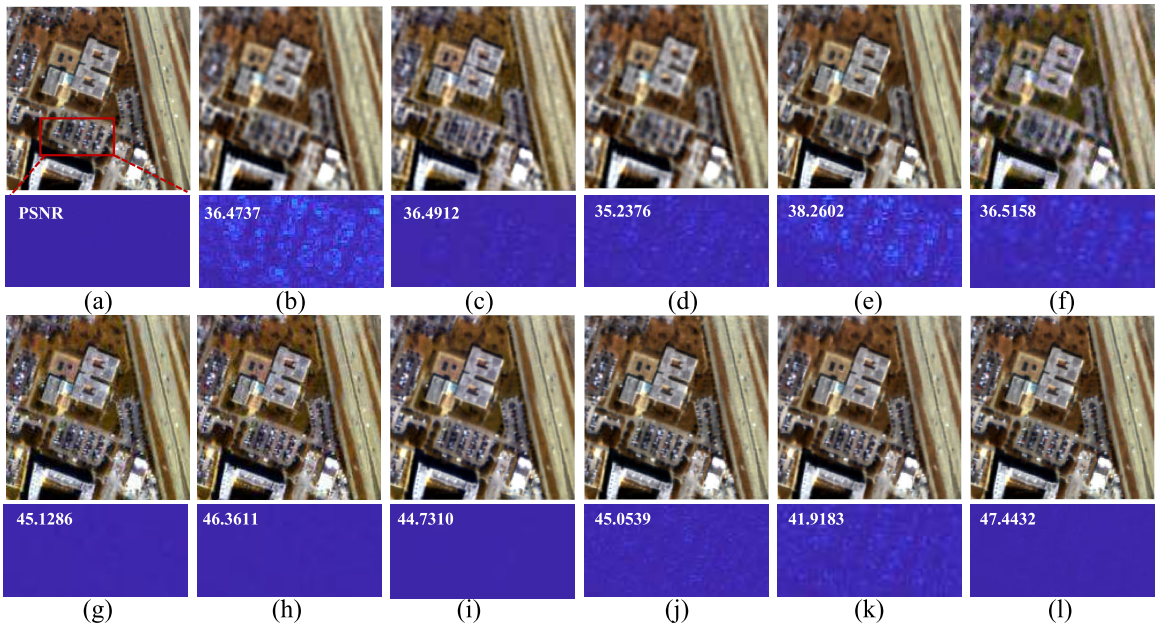


Fig. 6. Visual results and MAE maps generated by different HSI-SR algorithms for the unpaired Houston dataset. (a) Reference. (b) MSDFormer. (c) RepMamba. (d) CLSCNet. (e) Aware. (f) Aware (DMB). (g) Panmamba. (h) Panmamba (DMB). (i) MSAN. (j) DATSR. (k) RGTGAN. (l) Ours.

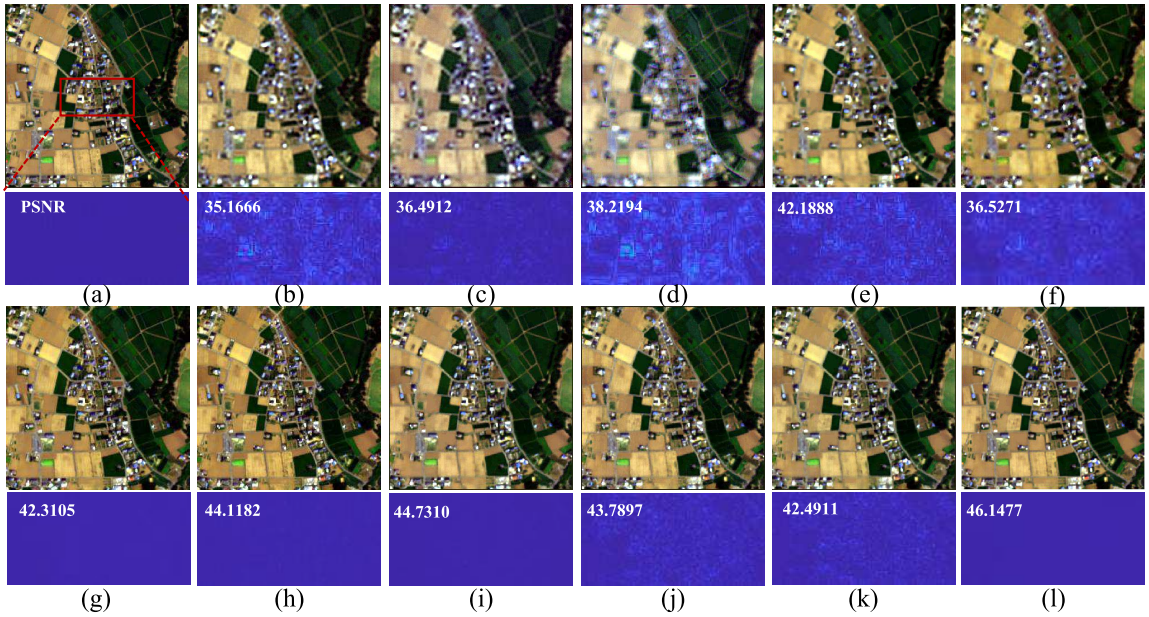


Fig. 7. Visual results and MAE maps generated by different HSI-SR algorithms for the unpaired ChikuseiR dataset. (a) Reference. (b) MSDFormer. (c) RepMamba. (d) CLSCNet. (e) Aware. (f) Aware (DMB). (g) Panmamba. (h) Panmamba (DMB). (i) MSAN. (j) DATSR. (k) RGTGAN. (l) Ours.

FIR DUO Rro R sensor with a spatial size of 700×350 pixels. The HSI was captured by a Micro-Hyperspec sensor covering a spectral range of 0.4–1.0 μm, comprising 270 spectral bands. After removing noisy bands, a total of 194 spectral bands were retained. Unlike other datasets, the Liyukou dataset employs a combination of multiple interpolation methods and Gaussian blurring to simulate comprehensive degradation in real-world scenarios.

B. Competing Methods and Quality Assessment

1) Competing Methods: Due to the lack of studies related to unpaired HSI-SR, our comparison methods give the following settings. The proposed method in this article addresses the problem of solving the lack of paired HrMSI for HSI-SR in practical applications, so the first comparison methods chosen are SISR methods that do not require HrMSI, such as MSDFormer [39], CLSNET [40], and RepMamba [41], and it is necessary to demonstrate that the proposed method outperforms the SISR method after the introduction of unpaired images. Second, we need to show that unpaired HrMSI makes the performance of fusion-based image SR degrade, so we chose Aware [42], Panmamba [18], and MSAN [43] as comparison methods. In addition, we compare the matched outputs of the DMB modules provided for these fused image SR methods, verifying whether DMB leads to an improvement

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in the performance of these methods. In addition, we also compare the case of providing DMB texture matching results for these fusion image SR methods to verify whether DMB leads to an improvement in the performance of these methods. Finally, we chose the Ref-SR method for natural images, DATSR [32], and the Ref-SR method for remote sensing images, RGTGAN [36], for comparison.

2) Quality Assessment: We have chosen five commonly used indicators to demonstrate the validity of the proposed method, including the spatial measure cross correlation (CC), the spectral measure spectral angle mapper (SAM), the global measure peak signal-to-noise ratio (PSNR), the root mean squared error (RMSE), and erreur relative globale adimensionnelle de synthese (ERGAS). The larger the value of CC and PSNR, the better, and their ideal values are 1 and positive infinity, respectively. The smaller the value of SAM, RMSE, and ERGAS, the better; their ideal values are all 0.

C. Experimental Setup and Results

1) Experimental Environment and Details: All experiments covered in this article were run on a device with Intel[1] Core[2] i9-10900X CPU @ 3.70 GHz and NVIDIA GTX 3090 GPU with OS Ubuntu 18.04.6 LTS. During training, we employed the Adam optimizer with a learning rate of $1e^{-4}$, a batch size of 4, and trained for 1000 epochs. All comparison experiments were trained and tested according to the configuration of the open-source code, and only the inputs and outputs of the models were changed to adapt them to the experimental data. Considering that spatial information of ground objects is required when calculating correlation during the matching process, patches need to be of a fixed size. Taking into account the size of ground objects and computational complexity, the proposed method uses patches of size 12×12.

In addition, the spatial degradation process in DMB is realized as an unsupervised pretrained network, and the latent spatial degradation has both fuzzy and downsampling effects. The spectral degradation is extracted from the parameters of the spectral camera, and in this article, it is assumed that this parameter is known at the imaging end of the HSI. Although the degradation processes of different datasets are different, the degradation targets are definite, and the two degradation processes are independent. Therefore, any unpaired HSI and MSI can be used as input. The dataset used in this article was constructed considering the similarity of unpaired image pairs.

2) Experimental Results: We conducted experiments on the Pavia, Houston, and ChikuseiR datasets, respectively, and the objective results of the experiments are displayed in Tables I–IV. As can be seen in Table II, on the constructed Pavia dataset, although the scenes contained in the HSI and MSI belong to the same style, there are still differences, and thus the objective metrics of this dataset are low compared to the traditionally paired Pavia Center dataset. Based on these results, the proposed model has the best performance, and the performance improvement of Aware and Panmamba with DMB applied is obvious. The Houston dataset represents a class of multitemporal unpaired datasets on which all methods work well due to the high similarity between image pairs as well as the simplicity of its content. Among them, Table II differs from Table I in that the performance of Aware after applying DMB is not improved, the reason may be that the network has limited ability to process texture features, and

[1]Registered trademark.
[2]Trademarked.

TABLE I
UNPAIRED HSI-SR RESULTS OBTAINED BY THE PROPOSED METHOD AND ITS COMPETING METHODS ON THE PAVIA DATASET. BOLD FONT IS BEST

Methods	CC	SAM	RMSE	ERGAS	PSNR
MSDFormer	0.9306	3.5178	0.0302	5.6705	30.5418
CLSCNet	0.9265	7.2219	0.0316	5.9522	30.1017
RepMamba	0.9530	5.7695	0.0241	4.6786	32.5091
Aware	0.9618	4.9306	0.0217	4.1875	33.4124
Aware(DMB)	0.9724	5.1266	0.0144	5.2368	36.5664
Panmamba	0.9843	3.7320	0.0126	2.7496	37.6112
Panmamba(DMB)	0.9855	2.9997	0.0114	2.2876	38.2368
MSAN	0.9828	4.0510	0.0131	2.7826	37.7255
DATSR	0.9848	3.8584	0.0119	2.5317	37.5920
RGTGAN	0.9760	4.4595	0.0161	3.2911	34.8989
Ours	**0.9888**	**2.7773**	**0.0099**	**2.1783**	**39.9066**

TABLE II
UNPAIRED HSI-SR RESULTS OBTAINED BY THE PROPOSED METHOD AND ITS COMPETING METHODS ON THE HOUSTON DATASET. BOLD FONT IS BEST

Methods	CC	SAM	RMSE	ERGAS	PSNR
MSDFormer	0.9703	2.7324	0.0167	3.0917	35.1666
CLSCNet	0.9927	1.6619	0.0076	1.4574	42.1888
RepMamba	0.9720	3.2127	0.0145	2.8148	36.4912
Aware	0.9836	2.2868	0.0118	2.2223	38.2194
Aware(DMB)	0.9768	3.5447	0.0164	3.2164	36.5271
Panmamba	0.9924	1.7161	0.0092	1.1484	42.3105
Panmamba(DMB)	0.9942	1.2298	0.0061	1.1484	44.1182
MSAN	0.9958	1.1750	0.0057	1.0837	44.7310
DATSR	0.9953	1.2320	0.0063	1.1856	43.7897
RGTGAN	0.9939	1.5244	0.0073	1.3657	42.4911
Ours	**0.9972**	**0.6444**	**0.0049**	**0.9155**	**46.1477**

TABLE III
UNPAIRED HSI-SR RESULTS OBTAINED BY THE PROPOSED METHOD AND ITS COMPETING METHODS ON THE CHIKUSEIR DATASET. BOLD FONT IS BEST

Methods	CC	SAM	RMSE	ERGAS	PSNR
MSDFormer	0.9407	2.1863	0.0137	5.0110	36.4737
CLSCNet	0.9104	2.7363	0.02043	6.8231	35.2376
RepMamba	0.9189	2.9102	0.0151	6.4434	35.8934
Aware	0.9598	2.0524	0.0111	4.2933	38.2602
Aware(DMB)	0.9259	3.3761	0.0124	9.5366	36.5158
Panmamba	0.9881	1.5338	0.0053	2.0163	45.1286
Panmamba(DMB)	0.9903	1.1647	0.0043	2.2351	46.3611
MSAN	0.9858	1.4008	0.0057	2.6369	44.2280
DATSR	0.9871	1.3168	0.0052	2.4561	45.0539
RGTGAN	0.9784	1.5438	0.0075	3.1213	41.9183
Ours	**0.9915**	**1.0559**	**0.0039**	**2.0502**	**47.4432**

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TABLE IV
UNPAIRED HSI-SR RESULTS OBTAINED BY THE PROPOSED METHOD AND ITS COMPETING METHODS ON THE LIYUKOU DATASET. BOLD FONT IS BEST

Methods	CC	SAM	RMSE	ERGAS	PSNR
MSDFormer	0.9905	1.3091	0.0049	1.1453	44.9904
CLSCNet	0.9798	2.4286	0.0101	2.3022	39.2870
RepMamba	0.9527	3.588	0.0122	2.6248	37.6554
Aware	0.9779	2.4262	0.0073	1.7736	41.5096
Aware(DMB)	0.9614	2.4510	0.0082	1.8799	41.1060
Panmamba	0.9918	1.1065	0.0042	0.8314	46.3155
Panmamba(DMB)	0.9936	0.9608	0.0037	0.7340	47.7587
MSAN	0.9951	0.7877	0.0035	0.7142	48.3353
DATSR	0.9939	0.9718	0.0037	0.7711	47.9124
RGTGAN	0.9850	1.1773	0.0059	1.1428	44.1241
Ours	**0.9949**	**0.78766**	**0.0034**	**0.7124**	**48.9651**

applying texture features without pairing the images with high similarity reduces the performance instead. The objective results for the Chikusei dataset are displayed in Table III, and consistent with the previous ones, the proposed method still has the best performance. Apart from this, we also conducted experiments on a real-world dataset. Unlike the simulated dataset, not only did our approach achieve optimal results, but several comparative methods also demonstrated commendable performance. This may be attributed to the limited resolution and reduced information content of the dataset we processed, which facilitated the application of multiple methodologies. This outcome underscores the broad practical applicability of theoretical approaches. From all the results, it can be seen that Panmamba, a method that uses Mamba structures to model the context to focus on global information, outperforms the other compared methods on the unpaired dataset, which is exactly why we chose to use Mamba in TCMM.

3) Visual Results: Figs. 5–8 show the visual results and the mean absolute error (MAE) map for the three datasets, respectively. We displayed a randomly selected image from the test set. The first image below each method name in the visual results is a pseudo-color image of the selected image, which selects three bands as the color image. The second image is the MAE map, which is obtained by taking the mean of the absolute value of the difference of all channels at each pixel point, so the more pronounced the texture of the MAE map, the greater the difference between the result and the reference image.

In conclusion, we conducted experiments on different types of unpaired datasets and found that when the scenes between HSI and MSI are more similar, the SR results are better.

D. Ablation Study

In this section, we have carried out some variants of the proposed method and conducted experiments on three datasets to confirm the effectiveness of the individual modules used in the proposed method.

1) Effect of Texture Matching: To confirm the effectiveness of the DMB, we conducted two ablation experiments, one to remove the branch of the MSI and the other to remove the DMB. The experimental results are shown in Fig. 9. From the results, it is evident that the performance of the network deteriorates when MSI is absent from the input, and that even when MSI is present but the textures do not match, it fails to enhance the performance of the network. This performance degradation stems from the lack of MSI branches, which prevents the network from extracting information directly from multimodal sources. The mismatched MSI data further exacerbates domain shifts in the LR, corrupting their spectral spatial characteristics and leading to worse results than the architecture that lacks MSI branching, compared to the architecture that retains MSI integration.

TABLE V
MODEL PARAMETERS AND FLOPS COMPARISONS FOR NETWORK STRUCTURES IN TCMM REPLACED WITH TRANSFORMER AND SS2D, RESPECTIVELY

	TCMM	Transformer	SS2D
FLOPs(G)	196.53	400.92	203.91
Parameters(M)	11.11	390.89	10.76

TABLE VI
COMPARATIVE EXPERIMENTAL RESULTS OF REGRESSION MODEL AND TRADITIONAL CONDITIONAL DIFFUSION MODEL

Method	Dataset	CC	SAM	RMSE	ERGAS	PSNR
Regression	Pavia	0.9872	3.5352	0.0104	2.2973	39.4677
	Houston	0.9965	0.9357	0.0054	1.0196	45.1135
	ChikuseiR	0.9904	1.1769	0.0039	2.1890	47.4190
Conditional DDPM	Pavia	0.9783	4.2830	0.0154	3.1018	36.2674
	Houston	0.9942	1.4025	0.0070	1.3198	42.8527
	ChikuseiR	0.9879	1.3197	0.0050	2.4150	45.4326
Ours	Paiva	0.9888	2.7773	0.0099	2.1783	39.9066
	Houston	0.9972	0.6444	0.0049	0.9155	46.1447
	ChikuseiR	0.9915	1.0559	0.0039	2.0502	47.4432

2) Effect of Texture Transfer and Enhancement: To demonstrate the performance of the proposed TCMM, we replace the structures in it with the transformer block and SS2D, respectively. Fig. 10 shows the results of the two ablation experiments, and Table V shows the computational complexity of each model. From the results, there is not much difference between the results of the three structures, but the unique structures of CIM and TEMamba in TCMM still achieved the optimal results. From the results, the proposed model achieves similar performance with less computational cost. Overall, the performance of variant-transformer is slightly lower than that of variant-SS2D, the reason may be the lack of the gating mechanism in the original transformer, which makes the texture transfer reasonably difficult. The vast majority of Ref-SR methods using the transformer structure have made some improvements to the transformer structure.

3) Effect of Diffusion Model: To validate the efficacy of the diffusion model, we conducted comprehensive ablation studies comparing three architectural variants: 1) the proposed CTMEM-Diff with iterative refinement; 2) a regression

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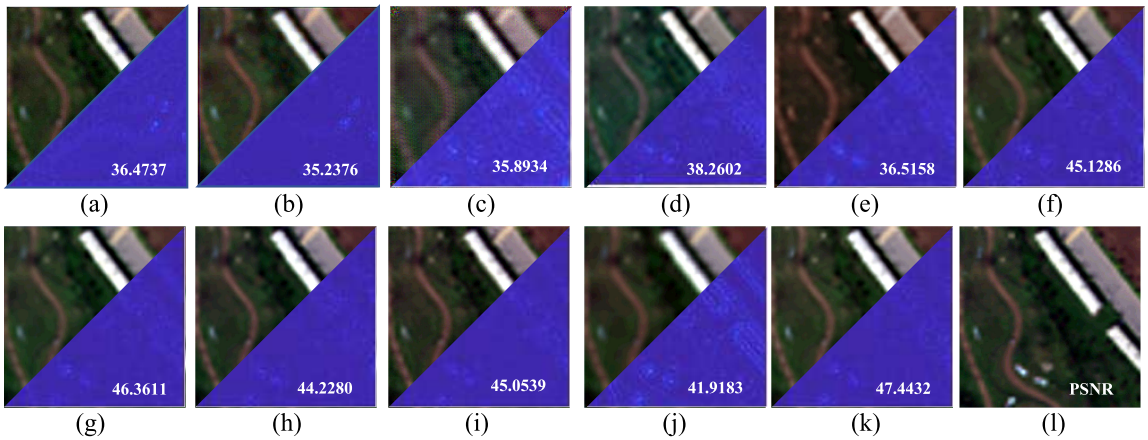


Fig. 8. Visual results and MAE maps generated by different HSI-SR algorithms for the unpaired Liyukou dataset. (a) MSDFormer. (b) RepMamba. (c) CLSCNet. (d) Aware. (e) Aware (DMB). (f) Panmamba. (g) Panmamba (DMB). (h) MSAN. (i) DATSR. (j) RGTGAN. (k) Ours. (l) Reference.

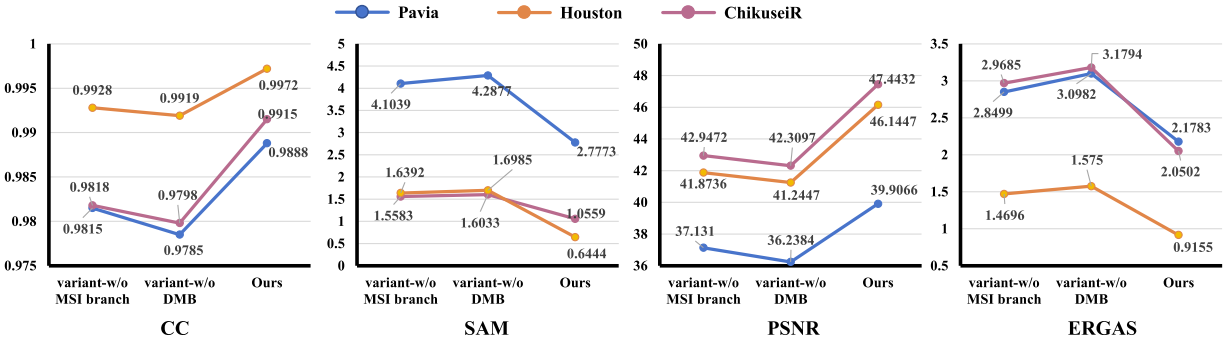


Fig. 9. Ablation results of the proposed method and its variants (MSI branch-ablated and DMB module-ablated) on the Pavia, Houston, and ChikuseiR datasets.

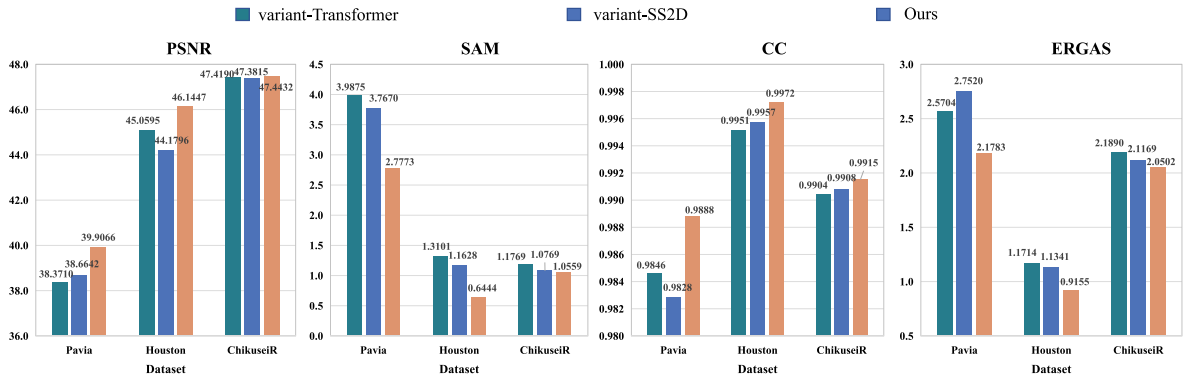


Fig. 10. Ablation study results of the proposed method and its transformer/SS2D variants on the Pavia, Houston, and Chikusei datasets.

baseline comprising only the DMB and TCMM modules; and 3) a conventional conditional diffusion model that concatenates the conditioning image with noisy inputs during the reverse process. To ensure experimental consistency, the conditional diffusion model followed the same experimental setup as our proposed model, with a learning rate of $1e^{-4}$, L1 loss, a total of 2000 iteration steps, and using the upsampled version of the HSI along with the corresponding MSI as the diffusion condition. The experimental results are presented in Table VI. As quantified in Table VI, the proposed method significantly outperforms the regression model, with an average PSNR improvement of 0.4981 dB and a SAM decrease of 1.1702 radians over the three datasets. Compared to the regression model, the proposed method also improves in two metrics, CC and ERGAS, which confirms that multiple iterations of the diffusion model can better refine the high-resolution details and effectively improve the quality of image generation. The conventional conditional diffusion model shows poor performance, with an average decrease of 2.9815 dB in PSNR, 0.0057 in CC, and an increase in SAM and ERGAS compared

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to the proposed method. This performance gap confirms our core hypothesis that simple concatenation cannot bridge the feature differences of features in unpaired images to establish effective cross-modal texture information transfer.

V. Conclusion

In this article, we propose a CTMEM-Diff model for the task of Unpaired HSI-SR, which gradually transfers and enhances the matched texture information in the framework of a diffusion model. In each step, the LrHSI and HrMSI are converted into a unified data dimension by different degradation functions in the DMB, which makes the process of texture matching more accurate. Subsequently, the matched texture images and the LrHSI and the noise image are fed into the TCMM to predict the desired HSI. We constructed three datasets based on five publicly available datasets and performed experimental tests on these three datasets. The experimental results showed that the proposed method achieved the best results among all the compared methods on five widely accepted objective indicators. In addition, through ablation experiments, we demonstrate the effectiveness of the DMB, TCMM, and diffusion processes used in CTMEM-Diff and the good performance of the proposed method with relatively low computational complexity.

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